

CLEARING THE WAY

Indian River Lagoon Stormwater Initiative

Complete Team Project Guide

5 Members · 1–3 Parent Volunteers · Summer 2026 Launch

West Shore Jr./Sr. High School · Melbourne, Florida · Brevard County

Primary Competition: CmPS (Community Problem Solvers) · Future Problem Solving Program International

About This Document This guide is written for a team of five students working together with 1–3 supporting parents. Every section assumes distributed workload, shared documentation, and coordinated scheduling across multiple families. Read the entire guide before beginning any field work. Follow the order of operations — especially the baseline water quality test, which must happen before any outreach begins. This document covers everything: field execution, competition strategy, documentation, scheduling, budgeting, and troubleshooting.

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PART 1 — PROJECT FOUNDATION

Section 1: Project Overview & Mission

What Clearing the Way Is

Clearing the Way is a student-led community awareness and environmental monitoring initiative targeting the Indian River Lagoon (IRL) in Brevard

County, Florida. The project operates on four parallel tracks: physically marking storm drains with permanent educational markers, educating households through two waves of direct-contact door hanger distribution, delivering classroom sessions to students in the drainage catchment area, and collecting scientific water quality data before, during, and after the campaign.

The core premise is straightforward: most residents in Melbourne do not know that the storm drains on their streets flow directly and untreated into the Indian River Lagoon. When fertilizer washes off a lawn during a rain event, when pet waste sits near a drain, when grass clippings blow into the street — all of that material travels through the storm drain system and into the lagoon within hours, with zero treatment and zero filtration. The lagoon receives it exactly as it left the street. Clearing the Way tells residents this fact, marks the drains so the reminder stays for years, and rigorously measures what changes as a result of the campaign.

This is not a cleanup project. It is not a one-day event. It is a precision behavior-change campaign targeting the specific residential habits — fertilizer application timing, pet waste management near drains, and debris removal from streets — that environmental science has identified as the primary controllable drivers of nitrogen and phosphorus loading in the Indian River Lagoon.

The Mission Statement

To protect the Indian River Lagoon by educating Brevard County residents about the direct connection between residential storm drains and the lagoon, permanently marking those drains as an ongoing visual reminder, and measuring the resulting changes in household awareness and water quality.

What Makes This Project Different

Most student environmental projects fall into one of several familiar patterns: they describe a problem, propose a theoretical solution, conduct a single cleanup event, or run a social media awareness campaign. Clearing the Way is

structurally different in four ways that matter to judges and to the lagoon itself.

Intervention at the source. The project targets storm drains — the entry points — rather than trying to remediate the lagoon after damage has occurred. Preventing one pound of nitrogen from entering the lagoon is orders of magnitude cheaper and more effective than removing it after it has fueled an algal bloom.

Built-in measurability. The pre- and post-survey structure, combined with water quality monitoring at two campaign sites and one control site across three rounds of testing, allows the team to present verified before-and-after evidence. Not anecdotal claims. Not projections. Actual field data with a control comparison built in. This level of scientific rigor is rare in student-led campaigns of any kind and distinguishes this project in every competition category it enters.

Designed for persistence. The drain markers last 3–5 years. The door hangers reach 300–500 households across two waves. The classroom sessions deposit knowledge with students who carry it home to their families. The city handoff at the end of the year means the drain database, the monitoring protocol, and the marking program all continue under city management after this team moves on. Judges across every competition ask: "What happens to this project when you leave?" Clearing the Way has a structural answer.

Team-based and documented. Five students with clearly divided roles, coordinated by documented team meetings, produces a richer CmPS portfolio than any undifferentiated group effort. Community Problem Solvers specifically rewards collaborative problem-solving, and this team's division of labor, meeting minutes, and coordinated execution will demonstrate that collaboration with concrete evidence.

Project Scope

Element	Team Target
Drains marked and GPS-logged	50–75
Households reached (both hanger waves)	300–500
Classroom sessions delivered	4–6
Students reached in classrooms	80–150
Water quality sites	3 (2 campaign zones + 1 control)
Water quality testing rounds	3 (baseline, mid-campaign, post-campaign)
Pre/post survey responses (target)	50+ (minimum 30 for statistical validity)
Partner letters obtained	At least 5 (city, 2 teachers, HOA, stormwater coordinator)

The One Sentence — Memorize This

"We marked storm drains across two Melbourne neighborhoods, educated the households above them with two rounds of direct outreach, brought the science into local classrooms, measured the water quality impact at three sites, and handed the complete program to the city so it continues after we graduate."

Every team member should be able to deliver this sentence fluently without looking at notes. It is the foundation of every application, every presentation, every conversation about the project. When a judge asks "So what did you do?" — this sentence is the answer. Everything else in your application is the evidence behind it.

Section 2: The Indian River Lagoon — Science, Stakes, and Story

Why the IRL Is the Right Target

The Indian River Lagoon is a 156-mile-long estuary running along the eastern coast of Florida through Brevard, Indian River, St. Lucie, Martin, and Palm Beach counties. It is a shallow, semi-enclosed coastal water body — meaning it does not flush efficiently. Water that enters it stays for extended periods, concentrating whatever pollutants it carries. This characteristic makes it especially vulnerable to nutrient loading from surrounding land.

The IRL contains more than 4,300 species of plants and animals, more species than any other estuary in North America. It supports 36 species listed as federally threatened or endangered. It is the only place in the world where Atlantic bottlenose dolphins, West Indian manatees, and right whales share the same waters. It has been designated one of the EPA's 28 National Estuary Programs — a recognition reserved for estuaries of national ecological and economic significance.

Since 2011, the lagoon has experienced recurring "superbloom" events — massive, system-wide cyanobacterial algal blooms that turn the water into an opaque green-brown mat for months at a time. The 2011 superbloom alone killed more than 60% of the lagoon's seagrass beds. That seagrass has not recovered. In the years since, these blooms have triggered mass die-offs of manatees, dolphins, and pelicans at rates that have alarmed federal wildlife managers and prompted emergency interventions. Brevard County estimates the cumulative economic impact of lagoon decline — through lost tourism, fishing, recreation, and property values — at approximately \$430 million per year.

The Nitrogen Connection — Why Stormwater Is the Problem

The primary driver of algal blooms in the IRL is excess nitrogen and phosphorus. These nutrients feed cyanobacteria and algae, allowing them to grow explosively when conditions are right. While there are multiple sources of nitrogen loading in the IRL — atmospheric deposition, septic systems, agricultural runoff — residential stormwater is one of the most significant controllable sources in urban Brevard County.

Here is the specific mechanism:

Fertilizer applied to residential lawns contains nitrogen in forms — primarily nitrate and ammonium — that dissolve readily in water. When rain falls within 48 hours of fertilizer application, or when irrigation water oversprays onto pavement, that nitrogen-rich water flows off the lawn, across the sidewalk, into the gutter, and directly into the nearest storm drain. From there it travels, without treatment, to the lagoon's edge and enters the water within hours.

Pet waste deposited near storm drains contains high concentrations of nitrogen and phosphorus, as well as fecal coliform bacteria. When it rains, pet waste dissolves into stormwater and travels the same path as fertilizer runoff — directly into the lagoon.

Grass clippings left on streets or sidewalks after mowing are nitrogen-rich organic matter. When they decompose in contact with rainwater, they leach nitrogen directly into stormwater and from there into the lagoon.

The critical distinction — the one that Clearing the Way communicates to every household it reaches — is this: **storm drains are not connected to the water treatment system. They flow directly to the lagoon.** Unlike water from your sink or toilet, which goes to a wastewater treatment plant before any discharge, stormwater receives no treatment at all. A resident who knows this fact may think differently about when to fertilize, where to let their dog go,

and whether to sweep clippings back onto the lawn. A resident who does not know this fact has no reason to change any of those behaviors.

The Public Health Dimension

The ecological damage is accompanied by documented public health consequences. Cyanobacterial blooms in the IRL produce cyanotoxins — microcystin, cylindrospermopsin, and anatoxin-a — that cause skin irritation, respiratory distress, and gastrointestinal illness in humans exposed through swimming, fishing, or contact with contaminated water or aerosols. The Florida Department of Health issues regular algal bloom advisories for IRL waterways in Brevard County, warning residents against water contact and seafood consumption during bloom events.

For HOSA Community Awareness judges in particular: Clearing the Way is interrupting the nutrient pathway at Step 1, before it ever enters the lagoon. Reducing the nitrogen and phosphorus that causes blooms reduces the toxin burden that creates the public health exposure pathway. This is a public health intervention that happens before anyone gets sick, which is the highest-value point to intervene.

The Numbers to Know Without Notes

Every team member should know these figures from memory. They appear in every competition application and every judge interaction.

Statistic	Value	Source
IRL length	156 miles	SJRWMD
Species count	4,300+	NOAA/EPA
Federally threatened/endangered species	36	USFWS

Statistic	Value	Source
Seagrass lost since 2011	60%	SJRWMD/FWC
Annual economic impact of decline	~\$430M/year	Brevard County
EPA National Estuary Program designation	Yes — 1 of 28	EPA NEP
Primary nutrient driver	Nitrogen & phosphorus from stormwater	SJRWMD research
Drain-to-lagoon travel time during rain event	Within hours	Melbourne Stormwater Division

Section 3: Team Structure & Roles

Why Role Division Matters for CmPS

Community Problem Solvers is evaluated on the process of collaborative problem-solving, not just the outcomes. Judges review how the team identified the problem, how they evaluated alternatives, how they divided responsibilities, and how they coordinated their work. A team that can document specific role contributions — this person managed the drain marking data, this person led the classroom sessions, this person compiled the survey analysis — scores higher on the implementation and collaboration rubrics than a team where responsibilities are undefined.

Assign roles before you begin any field work. Document the assignment in your first team meeting minutes. Stick to the roles throughout the project. This does not mean team members cannot help each other — it means each person has a defined primary responsibility and is the owner of that deliverable.

Recommended Role Assignments for a 5-Person Team

Role 1 — Project Lead / CmPS Report Lead - Faizan Ahmed

Responsibilities:

- Manages the overall project timeline and ensures milestones are hit on schedule
- Owns the CmPS report document — writes and compiles all sections with contributions from other roles
- Calls and runs team meetings; writes and distributes agendas; produces meeting minutes
- Coordinates with the FPS Florida coach on deadlines and registration requirements
- Drafts the city handoff package and coordinates the city meeting in February
- Leads the oral presentation at CmPS Florida Finals if the team advances

Key deliverables owned:

- CmPS Community Projects Proposal (all sections, compiled)
- CmPS Digital Portfolio structure
- Team meeting minutes archive
- City handoff cover memo and all city communication

Time investment: Slightly higher than other roles due to the coordination and writing load. Plan for approximately 20–25% more hours than the average role.

Best suited for: The team member with the strongest writing ability and organizational reliability.

Role 2 — Field Operations Lead (Drain Marking) - Kairav Kumar

Responsibilities:

- Manages all drain marking logistics: zone selection, session scheduling, supply obtainment
- Runs every drain marking session in the field
- Maintains the master drain GPS spreadsheet in real time at each drain
- Photographs every drain before and after marking; organizes the photo archive by drain ID
- Coordinates with Melbourne Public Works regarding the city sticker program
- Compiles the drain database export for the city handoff package

Key deliverables owned:

- Complete drain GPS database (Google Sheets, organized by drain ID)
- Before/after photo archive (named by drain ID and date)
- Drain marking session logs (date, drains completed, materials used, team present)
- Drain marker supply inventory and procurement records

Time investment: Heaviest during June–July (summer drain marking sprint). Significantly lighter during the school year once all drains are marked.

Best suited for: A team member who is comfortable outdoors, meticulous about physical records, and available for 4–6 Saturday morning sessions during the summer.

Role 3 — Community Outreach Lead (Door Hangers & Surveys)
Sankeerth Reddy Keisreddy

Responsibilities:

- Designs both waves of door hangers in Canva
- Sets up the pre-survey and post-survey Google Forms with correct questions
- Manages Wave 1 (July) and Wave 2 (November) hanger distribution logistics — street assignments, session scheduling, supply transport
- Tracks and maintains distribution logs: streets covered, hanger counts per street, dates and team members present
- Monitors survey response rates; deploys door-knocking if responses fall below target
- Analyzes pre/post survey data and writes the knowledge change summary for the CmPS report and competition applications

Key deliverables owned:

- Door hanger designs (Wave 1 and Wave 2 — different QR codes)
- Pre-survey and post-survey Google Forms
- Distribution logs for both waves (complete street-by-street records)
- Survey response dataset and knowledge change analysis

Time investment: Moderate and well-distributed — design work in June, distribution evenings in July and November, data analysis in December.

Best suited for: A team member with design interest or willingness to learn Canva, and comfort with direct community interaction.

Role 4 — Education Lead (Classroom Sessions) - Prithiv Janani's GF Ponnusamy

Responsibilities:

- Contacts science teachers and schedules all 4–6 classroom sessions

- Designs the 30-minute classroom presentation, the two-jar demonstration, and the take-home cards
- Delivers (or co-delivers with at least one other team member) all classroom sessions
- Collects teacher feedback letters at every session — on the spot, not later by email
- Documents all sessions: date, school, teacher name, attendance count, photo filenames, letter obtained

Key deliverables owned:

- Classroom presentation materials (slides or printed visual aid)
- Take-home card design and print order
- Teacher feedback/partnership letters (at least one per session)
- Classroom session documentation log

Time investment: Heaviest in September–October during scheduling and session delivery. Very light during summer.

Best suited for: A team member who is comfortable presenting to groups and has strong communication skills with adults.

Role 5 — Science & Data Lead (Water Quality Testing) - Aryan Pattu

Responsibilities:

- Selects both campaign water quality sites and the control site with GPS coordinates
- Purchases and maintains all water quality test kits
- Leads all three rounds of water quality field testing
- Maintains the water quality data spreadsheet with all parameters, all sites, all rounds
- Photographs all test sites each round from the same angle and position

- Writes the water quality analysis section of the CmPS report and other applications
- Compiles the water quality summary for the city handoff package
- Researches and cites published IRL nitrogen data to contextualize the team's field results

Key deliverables owned:

- Site selection documentation (GPS coordinates, photos, rationale)
- Complete water quality dataset (all parameters, all rounds, all sites)
- Field site photo archive (consistent angle, dated, captioned)
- Water quality summary with difference-in-differences framing

Time investment: Three concentrated field days (June, October, January) plus ongoing data management. Relatively low total hours but requires precision during active testing periods.

Best suited for: A team member with interest in environmental science, attention to detail with numeric data, and reliability when executing a specific protocol.

Team Meeting Structure

Hold a formal team meeting at minimum once per month. During summer (June–August), meet every two weeks.

Standing agenda for every meeting:

1. Status update from each role owner — 2 minutes each — what was completed, what is pending
2. Upcoming deadlines in the next 30 days
3. Any blockers or problems to solve as a group
4. Decisions requiring full team agreement

5. Action items for each person with specific due dates

After every meeting: The Project Lead writes a one-page summary — date, attendees, key discussion points, decisions made, action items by person with deadlines. This goes into the shared Google Drive folder labeled "Team Meeting Minutes." These minutes are evidence for the CmPS portfolio and demonstrate collaborative process.

Parent attendance: At least one parent should attend the first meeting and any meeting involving a financial decision, a new institutional partnership, or a major logistics commitment. Parents do not need to attend routine working meetings.

Section 4: Parent Involvement Guide

Overview

Parent involvement in Clearing the Way is logistical, not academic. Parents are not expected to understand nitrogen loading, the CmPS rubric, or the Earth Prize Fresh Idea format. They are expected to be available for transportation, safety supervision at water sites, and occasional administrative support. Every parent task below has a specific, bounded time estimate.

What Each Parent Task Actually Involves

Transportation — Drain Marking Sessions (Summer)

The team conducts 6–8 drain marking sessions on Saturday mornings in June and July, each lasting approximately 2–3 hours. A parent vehicle is needed to transport team members to the campaign zone and to carry marking supplies. Parents do not need to watch the marking process — they can run nearby errands, sit in the car, or work in the area.

Parent time per session: 2–3 hours. **Total across all sessions:** 12–18 hours distributed across the parent pool.

Safety Supervision — Water Testing Sites (3 Rounds)

All three water quality testing rounds require at least one adult present at the water's edge. This is a non-negotiable safety requirement — students should not be at waterway edges without adult supervision. The adult does not need to operate the test kits.

Parent time per round: 2–3 hours (travel to both sites, presence during testing, return). **Total across all rounds:** 6–9 hours.

Transportation — Door Hanger Distribution (Summer and Fall)

Wave 1 (July) and Wave 2 (November) each involve 3 distribution evenings of approximately 2 hours each. Parents transport the team to the campaign zone and wait nearby.

Parent time per distribution evening: 2–2.5 hours. **Total for both waves:** 12–15 hours distributed across the parent pool.

Transportation — Classroom Sessions (September–November)

Each classroom session requires transportation to and from a school. Some schools may require a parent to sign in as a supervising adult. Parents do not present during sessions.

Parent time per session: 1.5–2.5 hours (varies by school distance). **Total for 4–6 sessions:** 6–12 hours distributed.

City and HOA Meetings (February)

The city handoff meeting and the HOA presentation each require parent transportation and a supportive adult presence in the room. Students lead both meetings.

Parent time: Approximately 2–3 hours per meeting.

Full-Year Parent Time Summary

Task	Estimated Hours (Total Across Parent Pool)
Drain marking transportation	12–18 hrs
Water testing supervision	6–9 hrs
Door hanger distribution transportation	12–15 hrs
Classroom session transportation	6–12 hrs
City and HOA meetings	4–6 hrs
TOTAL	~40–60 hrs total (distributed across 1–3 parents)

With three parents sharing this load: approximately 15–20 hours each over the full year. With two parents: approximately 20–30 hours each. These are manageable, plannable commitments — not daily involvement.

Parent Communication Protocol

Assign the Project Lead to send a brief monthly update to all participating parents — no longer than one paragraph, answering three questions: what happened this month, what is coming next month, and what specific parent involvement is needed and when. Create a group text or messaging thread for day-of logistics only (reminders, schedule changes, cancellations). Keep planning conversations among the student team.

Section 5: The Four-Prong Model

Overview

Clearing the Way addresses residential stormwater pollution through four coordinated approaches. They are called prongs because they converge on the same target from different directions. No single prong is sufficient alone. Together they create overlapping exposure — a resident might encounter the project through a marked drain, a door hanger, a take-home card from their child, and water quality news shared at an HOA meeting. Each additional point of contact reinforces the behavior-change message.

Why Four Prongs

The research on behavior-change campaigns is consistent: single-channel interventions produce small and temporary effects. People need repeated exposure through multiple modalities before behavior changes durably. The marked drain is a persistent visual cue every time it rains. The door hanger delivers specific, actionable information to the household decision-maker. The classroom session reaches children who bring what they learned home. The water quality data creates scientific accountability.

For CmPS specifically: the four-prong model provides the foundation for the alternatives analysis (the team evaluated single-channel approaches and documented why each was insufficient) and the best-solution justification

(each prong addresses the behavior change problem through a complementary mechanism). These are the sections CmPS international judges score most carefully.

Prong 1: Mark

What it is: GPS-logging and permanently marking 50–75 storm drains in two residential campaign zones. Each drain gets an educational marker ("DRAINS TO LAGOON"), is photographed before and after marking, and is logged with GPS coordinates and a unique drain ID.

Behavior-change mechanism: The marker creates a persistent visual cue at the exact moment and location of potential harm — when a resident approaches a drain with a lawn mower, fertilizer spreader, or dog on a leash, the marker communicates the consequence of what they're about to do. Research on environmental cue placement consistently shows that visual reminders at the point of decision are more effective than general awareness campaigns.

Persistence: Vinyl stickers and paint stencils last 3–5 years. The city handoff means the municipality refreshes markers on an ongoing basis.

Prong 2: Inform

What it is: Two waves of door hanger distribution to 300–500 households. Wave 1 (July) includes a pre-survey QR code at the top of the folded hanger, visible before any content is read. Wave 2 (November) replaces the pre-survey QR with a post-survey QR.

Behavior-change mechanism: The hanger reaches the adult household decision-maker — the person who controls fertilizer application, pet management, and lawn debris. The two-wave structure provides both the educational message and the measurement tool in one delivery system.

The QR placement rule: The pre-survey QR must appear at the very top of the folded panel, before any educational content. This ensures the survey captures knowledge before any educational exposure occurs — which is the methodological requirement for a valid pre/post comparison. This design decision must be explained in every competition application.

Prong 3: Educate

What it is: Four to six 30-minute classroom presentations at schools in or near the campaign drainage catchment. Each session includes a two-jar demonstration, interactive behavior-change content, and take-home cards personalized with each student's name.

Behavior-change mechanism: Students are messengers to their households. A student who learns in class that their street drains directly to the lagoon will talk about it at home. Take-home cards personalized with the student's name increase parent read rates significantly. This prong also reaches a fundamentally different audience than the door hanger — renters, families who missed the hanger, and younger household members who may influence adult behavior.

Prong 4: Measure

What it is: Three rounds of water quality testing at three sites — two campaign zone stormwater outfalls and one control site. Parameters tested: nitrate, phosphate, pH, temperature, and turbidity. Testing occurs before outreach (baseline), during the campaign (mid-point), and after all outreach is complete (post-campaign).

Purpose: The water quality data provides independently verifiable scientific evidence of campaign effect. Most community awareness campaigns have no

independently measurable environmental outcome. Clearing the Way has both survey-based behavioral evidence and water quality field data. Even if the water quality data shows no statistically clear improvement — which is scientifically expected in a single year — the presence of a rigorous monitoring protocol with a control comparison signals scientific credibility that other student campaigns lack.

PART 2 — FIELD WORK

Section 6: Prong 1 — Storm Drain Marking (50–75 Drains)

Why 50–75 Drains

Fifty to seventy-five drains across two campaign zones is achievable in 6–8 Saturday sessions for a five-person team. It represents meaningful physical coverage — typically 4–8 residential streets per zone — while staying within the documentation capacity the team can maintain with high quality.

Documentation quality is what judges score, not raw drain count. Fifty well-documented drains with GPS coordinates, timestamped before/after photos, and a clean spreadsheet record are more valuable in a competition portfolio than 100 drains with gaps and inconsistencies.

The two-zone structure provides a richer comparison for water quality data: Campaign Site A and Campaign Site B each have their own outfall for testing, with one control site in a separate neighborhood. This three-site design is more defensible scientifically than a single campaign zone.

Step 1: Zone Selection

Email Melbourne Public Works at publicworks@melbourneflorida.org:

Subject: Student stormwater drain project — request for drainage maps

Hi, we are West Shore High School students working on a community stormwater awareness project focused on the Indian River Lagoon. Could we get copies of the stormwater drainage maps for [neighborhood name] and [second neighborhood name]? We are identifying which residential streets drain to the lagoon through specific outfalls. This is for a school environmental project. Thank you.

This is a public records request. Most cities provide drainage maps free within one week. If there is no response within 7 business days, call directly: (321) 674-5752.

From the maps:

1. Find the outfall point — where the storm drain system exits into the lagoon or a tributary. This becomes Campaign Site A or B for water quality testing.
2. Trace backward from the outfall to identify which streets drain into it. These are your campaign zone streets.
3. Count drains on those streets using Google Street View before the first physical visit. Aim for 25–40 drains per zone.
4. If more than 40 drains appear in a zone, select the streets with the highest residential density — most houses above the drains.

Step 2: Pre-Session Preparation

Before each drain marking session, the Field Operations Lead prepares:

- Drain ID assignment list (sequential numbers, e.g. Drain-001 through Drain-075, not restarting for Zone B)
- Google Sheet open and ready on phone for real-time logging
- All materials packed and checked: stickers or stencil + paint, alcohol wipes, gloves, smartphone
- Photo naming convention reviewed with any team member new to the session

Step 3: GPS Logging Each Drain (2–3 Minutes Per Drain)

At each drain, complete these steps before moving to the next one:

Step	Action	Record
1	Stand over the drain. Open Google Maps. Press and hold your exact location. Copy the GPS coordinates.	Drain_ID, GPS_Lat, GPS_Lon in Google Sheet
2	Note nearest house address	Street_Address column
3	Take "before" photo: 4 feet back, full drain cover + one identifiable landmark (house number, street sign, utility pole)	Name file: YYYYMMDD_drain_001_before.jpg
4	Apply marker (see Step 4 below)	
5	Take "after" photo from same position	Name file: YYYYMMDD_drain_001_after.jpg

Step	Action	Record
6	Open Google Sheet and add one complete row	Date, DrainID, GPS, Address, Material, Photo filenames, Team member initials

Never batch-log at the end of a session. Logging at each drain takes 90 seconds. Logging from memory at session end takes longer and produces errors you cannot fix later.

Step 4: Marking Each Drain (3–5 Minutes Per Drain)

Option A — City "Drains to Lagoon" Stickers (Preferred)

Request these in your first email to Melbourne Public Works. Many Florida municipalities provide them free for community campaigns. City-provided stickers carry the city's visual identity, which strengthens the partnership documentation for the handoff.

Application: Wipe drain surface with alcohol wipe. Allow 30 seconds to dry. Peel sticker backing from center outward. Press firmly from center out for 30 seconds. Do not apply in rain or on wet surfaces.

Option B — Custom Vinyl Stickers

Order from StickerMule.com: round weatherproof vinyl stickers, 3-inch diameter. Design: "DRAINS TO LAGOON" in bold type, a simple wave or droplet icon, project website URL. Order 100 minimum. Cost: approximately \$25–30 for 50 stickers.

Application: Same as city stickers.

Option C — Paint Stencil (Fallback)

Purchase an aluminum or durable plastic stencil with "DRAINS TO LAGOON" cut out (Amazon, \$15–25). Use Rust-Oleum spray paint in blue or teal (\$6/can).

Application: Place stencil flat on drain cover. Hold firmly. One short spray burst from 6–8 inches above. Hold stencil in place 15 seconds before lifting. Do not spray in wind.

Permission note: Check with Melbourne Public Works whether stenciling storm drains requires written authorization in your specific area. City stickers and city-approved vinyl stickers are universally permitted. Stenciling may require a brief email authorization from the stormwater division — request it proactively.

Step 5: Session Logging

At end of each session, the Field Operations Lead logs in a separate "Session Log" tab:

- Session date and time (start and end)
- Team members present
- Drains completed (ID range)
- Zone worked
- Weather conditions
- Any drains skipped and reason
- Materials used and remaining supply count

Marking Schedule

Session	Drains	Zone	Timing
1	001–012	A	Late June, Saturday AM

Session	Drains	Zone	Timing
2	013–025	A	Early July, Saturday AM
3	026–038	A	Mid-July, Saturday AM
4	039–050	B	Late July, Saturday AM
5	051–062	B	Early August, Saturday AM
6	063–075	B	Mid-August if needed

With 5 team members averaging 3–5 minutes per drain (GPS + photo + mark + log), the team completes 12–15 drains per 2-hour session. Six sessions = 72–90 drains — more than enough to hit the 50–75 target even with skips.

Section 7: Prong 2 — Door Hanger Campaign (300–500 Households)

Overview

Two waves of door hangers reach every residential household above the marked drains. Wave 1 in July carries the pre-survey QR code; Wave 2 in November carries the post-survey QR code. The knowledge change measured between the two surveys — specifically the percentage of respondents who correctly identify that storm drains flow directly to a waterway — is the project's primary quantitative outcome.

Set Up the Survey Before Printing Anything

The pre-survey must be live and the QR code must be embedded in the hanger before a single hanger is distributed. Design the hanger and set up the survey simultaneously in June Week 3.

Platform: Google Forms (free). Responses go automatically to a linked Google Sheet. Generate the QR code at qr-code-generator.com (free).

Survey questions — use identical wording in both pre and post surveys:

Q#	Question	Options
Q1 (PRIMARY)	When it rains, where does the water from your street go?	(a) To a water treatment plant (b) Directly to a waterway without any treatment (c) It soaks into the ground (d) I don't know
Q2	How concerned are you about the Indian River Lagoon's health?	Scale 1 (not concerned) to 5 (very concerned)
Q3	How often have you applied fertilizer to your lawn in the past 12 months?	Never / 1–2 times / 3+ times / I don't have a lawn
Q4	When walking a dog, how often do you pick up waste within 100 feet of a storm drain?	Always / Usually / Sometimes / Never / I don't have a dog
Q5 (Wave 2 only)	Have you seen the "Drains to Lagoon"	Yes / No / Not sure

Q#	Question	Options
	markers on storm drains in your neighborhood?	

Q1 correct answer: (b). The before-vs-after percentage selecting (b) is your headline statistic in every application.

Minimum responses: 30 per wave for statistical credibility. Target 50+. If below 15 after two weeks, deploy the door-knocking protocol in Section 20.

Designing the Hanger in Canva

Go to canva.com — free account. Search "door hanger." Select a template with the hanging hole at top and a fold line below it.

Front panel layout (from top to bottom):

1. Pre-survey QR code + "Scan before reading — 60 seconds" instruction
2. Project headline: "Your street drains directly to the Indian River Lagoon."
3. Simple graphic: house → street → drain → lagoon (no treatment step)
4. Project name and website URL

Back panel layout:

1. "Three things you can do right now"
2. Message 1: Fertilize only when no rain is forecast for 48 hours
3. Message 2: Pick up pet waste before it rains, especially near drains
4. Message 3: Sweep grass clippings back onto the lawn — never into the street
5. Footer: Project website, contact information

Print quantity: Target number of households plus 10% overage. For 400 households, print 440. Cost: approximately \$45–60 per wave at a local print shop. Ask for a student project or nonprofit discount.

Distribution Protocol

With five team members, each evening session covers 2 hours of walking. Pre-assign each member 1–2 specific streets before leaving the car — no ad-hoc division in the field.

Per session:

1. Each member counts hangers placed on their assigned streets (tally on phone notes app)
2. Place one hanger per door handle or mailbox. Never skip a house.
3. If a resident comes to the door: "Hi, we're West Shore students working on a lagoon protection project — leaving information for every household on this street. There's a quick survey on top if you have a minute."
4. When returning to the car, immediately record: streets covered, hanger count per street.
5. That evening, before sleep: log the full session — date, team members, streets, total hangers, cumulative total.

Three evenings per wave × 80–120 households per 2-hour evening = 240–360 per wave. This meets the 300–500 household target with two waves.

Wave 2

Wave 2 goes to the exact same streets in November — approximately 4 months after Wave 1. The only design change is the QR code at the top: replace the pre-survey code with the post-survey code. All other content stays the same.

In December, after Wave 2 response collection is complete, the Community Outreach Lead produces a one-page analysis:

- Total responses: Wave 1 (pre) and Wave 2 (post)
 - Q1 correct-answer rate before vs. after
 - Q2 average concern before vs. after
 - Q3 and Q4 behavior frequency before vs. after
 - Q5: percentage who report seeing drain markers
-

Section 8: Prong 3 — Classroom Sessions (4–6 Sessions)

Overview

Four to six classroom sessions at schools within or near the drainage catchment is the right target for a five-person team. The Education Lead schedules and leads each session; at least one other team member attends each one for documentation support and to demonstrate a team-based initiative.

How to Get Sessions Scheduled

Target schools first: West Shore Jr./Sr. High School is the easiest first ask — team members have existing relationships with their own science teachers.

Email to science department chair:

Subject: 30-minute lagoon watershed presentation for your class — NGSS aligned

Hi [Teacher name],

We are a team of West Shore students running a community stormwater protection project focused on the Indian River Lagoon.

We have a 30-minute classroom presentation on how residential storm drains connect directly to the lagoon, including a hands-on water demonstration and take-home cards for students' families.

The session aligns with NGSS MS-ESS3-3 and HS-ESS3-4. It includes interactive discussion and three specific, actionable behavior-change messages for households.

Would you have 30 minutes in September or October? We are fully flexible on timing and can work around your schedule. We can also leave extra materials for any colleagues who might be interested.

Thank you, [Team member names], Clearing the Way Initiative, West Shore Jr./Sr. High School

Follow up once by email if no response within one week. If still no response, call the school's main office and ask to speak with the science department.

If a teacher declines, ask for a colleague referral — a referral doubles your acceptance rate with the referred teacher.

The 30-Minute Session Script

0:00–3:00 — Opening

Three show-of-hands questions. Pause after each. Do not give answers yet.

1. "Who has been to the Indian River Lagoon — swimming, fishing, kayaking, anything?"
2. "Who lives within about a mile of the lagoon?"
3. "Who can tell me where the water from your street goes when it rains?"

Accept any answers without correcting. Build curiosity: "Let me show you what we found."

3:00–8:00 — The Core Reveal

Show one Canva graphic: house → street → storm drain → pipe → lagoon, with a clear X over any treatment step.

Say: "Unlike your sink or toilet — which go to a water treatment plant before any discharge — storm drains go directly to the lagoon. No filter. No treatment. Whatever is on your street gets to the lagoon within a few hours of any rain event."

Let it land. Five seconds of silence. Then: "That's the whole project. Everything we do is based on that one fact, because most people don't know it."

8:00–18:00 — Two-Jar Demonstration

Materials (Education Lead brings to every session): two large mason jars, one large clear bowl or container, blue food coloring, green food coloring, potting soil (1 tsp), cooking oil (1 drop), one blade of grass.

Prepare before class: Jar A = water + blue food coloring. Jar B = water + green coloring + soil + oil + grass.

At 8 minutes: "Jar A is water from a street where residents fertilize carefully, pick up pet waste, and sweep their clippings back onto the lawn. Jar B is water from a typical residential street without those habits."

Pour both jars simultaneously into the large container. "Which one would you swim in? Because this — right here — is what the lagoon gets during a rain event on a typical residential street."

Ten seconds of silence. The visual does the work.

18:00–24:00 — Interactive Behavior Change

Ask: "What could households do differently? Call out three things." Write answers on the board as students generate them. Guide toward:

1. Fertilizer timing — wait 48 hours after fertilizer before rain, or use slow-release
2. Pet waste near drains — pick up before it rains
3. Grass clippings — sweep back onto the lawn, never into the street

Having students generate the answers produces better retention than lecturing. They own what they said.

24:00–28:00 — Take-Home Cards

Distribute cards. Have each student write their first name on the "Brought to you by [Name]" line before taking it.

"This goes to whoever takes care of your yard at home. You now know something about the lagoon that most adults in Melbourne don't know. Tell them."

The personalization line is not decorative — parents read cards with their child's name significantly more than generic informational materials.

28:00–30:00 — Wrap-Up

Q&A for two minutes. Thank the teacher by name in front of the class. Then:

"Before we go — [Teacher name], could you fill out this quick form confirming what we covered today? It just takes about two minutes and it's just your name, the date, the class size, and a signature."

Hand the teacher the pre-written template immediately. 80% sign on the spot. If you leave without the signature and email later, the response rate drops dramatically.

Documentation at Every Session

Before leaving the building:

- **Photo:** One photo of the class (with teacher permission), one of the jar demonstration. Caption immediately.
- **Attendance:** Ask the teacher the approximate count.
- **Teacher letter:** Obtained on the spot.
- **Card count:** Count cards distributed.

Log the session that evening: date, school, teacher name and contact, class level, approximate attendance, photos logged, letter obtained (Y/N).

Section 9: Prong 4 — Water Quality Testing (3 Rounds, 3 Sites)

The Three Sites

Site	Location	Purpose
Campaign Site A (CS-A)	Stormwater outfall from Zone A campaign streets — on public land	Measures receiving water quality for Zone A stormwater
Campaign Site B (CS-B)	Stormwater outfall from Zone B campaign streets — on public land	Measures receiving water quality for Zone B stormwater

Site	Location	Purpose
Control Site (CTR)	Outfall in a comparable residential neighborhood NOT in either campaign zone — at least 0.5 miles away	Captures seasonal variation; enables difference-in-differences comparison

Identifying campaign sites: Use the drainage maps from Melbourne Public Works. The outfall is where the drain pipe exits into a canal, tributary, or the lagoon directly.

Identifying the control site: After finalizing campaign zones, select a neighborhood of similar residential density and land use, not adjacent to either campaign zone, with an accessible outfall on public land.

GPS every site exactly. Identify a permanent landmark at each site for photo anchoring — you return to the same spot three times and need photo-comparable frames.

Testing Schedule

Round	When	Condition
Round 1 — Baseline	June 15–17, 2026	Before ANY outreach. Non-negotiable. Doing outreach first destroys the baseline.
Round 2 — Mid-campaign	Late October 2026	After Wave 1 hangers and drain marking; before Wave 2

Round	When	Condition
Round 3 — Post-campaign	Late January 2027	After Wave 2 hangers, post-survey, and classroom sessions

What to Test

Parameter	Kit	Cost	Why
Nitrate	API Freshwater Master Test Kit	\$35 (all 3 rounds, all 3 sites)	Primary lagoon nutrient driver
Phosphate	LaMotte phosphate test strips	\$18 (90-strip bottle)	Secondary nutrient; pet waste connection
pH	Included in API kit	Already included	Contextualizes other measurements
Temperature	Waterproof digital thermometer	\$10	Required for all parameter interpretation
Turbidity	Visual observation only	Free	Proxy for algal growth and sediment load

Total kit cost: ~\$63. No expensive equipment needed.

Field Protocol — 20 Minutes Per Site

Perform in this exact order at every site every round:

1. **Photograph the site** from the same position and angle as prior rounds, with the permanent landmark visible.
2. **Record conditions:** Date, time, weather, visible algae or discoloration, rainfall in prior 48 hours (check weather.gov for Melbourne FL).
3. **Rinse the sample bottle** with site water before filling.
4. **Collect the sample** 6–12 inches below surface, 1–2 feet from the outfall mouth. Cap immediately.
5. **Run all tests within 2 hours** of collection. Log every result before leaving the site.
6. **Photograph the data sheet** as a backup record.

Water Quality Spreadsheet Structure

Column	Values
Round	1, 2, or 3
Site	CS-A, CS-B, or CTR
Date	YYYY-MM-DD
Time	HH:MM
Weather	Clear / Cloudy / Rain
Prior_48hr_Rain	Yes / No / Trace
Turbidity_Visual	Clear / Slightly murky / Visibly murky / Opaque
Nitrate_mgL	Numeric
Phosphate_mgL	Numeric
pH	Numeric
Temperature_C	Numeric
Tester_Initials	Who ran the tests

Column	Values
Notes	Anomalies, site conditions

Reporting Results Honestly

If campaign sites improve more than control: Report as a possible campaign-adjacent signal with the difference-in-differences framing. Lead with the specific number.

If all sites change similarly: Report as regional seasonal variation driving both sites; note that pre/post survey provides independent behavioral evidence.

If no change anywhere: Report as consistent with documented lag times in estuarine nutrient response; note that 6–9 month monitoring periods are routinely insufficient to detect behavioral-change-driven improvements at estuary scale. Year 2 monitoring extends the longitudinal baseline.

All three outcomes are scientifically valid and acceptable to every judge in every competition. Honesty is more credible than any stretched positive claim.

Section 10: The Documentation System

The Core Rule

Document at the site, before you leave. Every time. Not that evening. Not this weekend. At the site, before getting back in the car. This rule applies to every team member in every field activity. There are no exceptions. Documentation produced from memory is documentation with errors you cannot identify.

Three-Platform Architecture

Platform	What Goes Here	Who Has Access
Google Sheets — Master Project Log	All field data: drain log, distribution log, session log, water quality data, survey exports	All team members
Google Photos — "Clearing the Way 2026"	All field photos, renamed before upload	All team members
Google Drive — "Clearing the Way Documents"	Partner letters (scans), meeting minutes, competition drafts, handoff package documents	All team members

Google Sheets tabs:

- **Drains:** One row per drain — Drain_ID, Zone, Date_Marked, GPS_Lat, GPS_Lon, Street_Address, Material, Before_Photo, After_Photo, Team_Member
- **Distribution:** One row per session — Wave, Date, Streets_Covered, Hanger_Count, Team_Present, Notes
- **Sessions:** One row per classroom session — Date, School, Teacher, Grade, Attendance, Cards_Distributed, Photos, Letter_Obtained
- **Water Quality:** One row per site per round (full structure in Section 9)
- **Survey Data:** Monthly exports from Google Forms

Photo Naming Convention

Name every photo before uploading:

YYYYMMDD_[type]_[identifier].jpg

Examples:

20260615_wq_cs-a_round1.jpg

20260712_drain_017_before.jpg

20260712_drain_017_after.jpg

20260903_classroom_westshore_chen.jpg

20261025_wq_ctr_round2.jpg

20261105_distribution_wave2_palmave.jpg

Partner Letters

You need a minimum of five external partner letters. Bring a pre-filled template to every partner interaction. Hand it to the person at the end of your meeting: "This just confirms what we did together — two minutes to fill in and sign." People who receive a template and are asked to sign immediately do so at a much higher rate than people asked to "send a letter later."

Source	Confirms
Melbourne Stormwater Coordinator	Drain marking partnership; handoff receipt
Teacher 1	Date, school, attendance, activity description
Teacher 2	Same
HOA President	HOA meeting; community partnership

Source	Confirms
MRC LagoonWatch (if training completed)	Methodology training; scientific credibility

PART 3 — COMPETITION STRATEGY

Section 11: CmPS — Primary Competition (Deep Dive)

What CmPS Is

The Community Problem Solvers component of Future Problem Solving Program International (FPSP) is a year-long project-based competition where student teams identify a real community problem, systematically analyze alternatives, implement the best solution, and document measurable results. Projects advance from the school level to Florida Affiliate Finals and potentially to International Conference.

CmPS is the right primary competition because the project was designed with the CmPS rubric in mind. The four-prong model directly satisfies the rubric's requirements: documented problem identification, alternatives analysis, implementation evidence, measurable community impact, and a sustainability plan.

Registration and Key Dates

Milestone	Date
FPS Florida registration opens	July 1, 2026 — register the day it opens
Community Projects Proposal due	~February 27, 2027 — verify with FPS Florida coach
Digital Portfolio + Video due	~March 22, 2027
Florida Affiliate Finals (in-person)	~March 23–26, 2027 — location TBD
International Conference	~June 2027 — only if advancing from FL Finals

H4 status: Open. FPS does not screen for immigration status. Florida school enrollment is the only requirement.

The CmPS Evaluation Rubric — Section by Section

Section 1: Problem Identification (20 points)

Judges score specificity, precision, and data support. Not a general environmental statement — a specific, bounded, evidence-based problem definition.

Wrong: "The Indian River Lagoon is polluted and needs help."

Right: "Residential nitrogen loading through unmarked stormwater outfalls is the primary controllable driver of cyanobacterial blooms in the IRL's Brevard County segment. Pre-campaign survey data shows that [X]% of households above the study zone's storm drains were unable to identify that those drains flow directly to a waterway without treatment — a quantified information gap that functions as the primary behavioral lever for intervention."

What the team provides: published IRL nitrogen data (SJRWMD, EPA NEP), pre-survey Q1 response data, a specific geographic problem definition, and a clear causal chain: residential behavior → stormwater nitrogen → algal blooms → seagrass loss → economic and ecological damage.

Owner: Project Lead, with Community Outreach Lead (pre-survey data) and Science Lead (published data).

Section 2: Alternatives Analysis (20 points)

Judges score the quality of the team's decision-making process. Not just what they chose — but why they chose it over real alternatives, using consistent criteria.

Alternatives table:

Alternative	Cost	Behavior Change Mechanism	Measurability	Persistence	Feasibility for Team	Score
Social media campaign	Low	Indirect; skips older demographics	Low	Low	High	7/25
City council advocacy	Low	Systemic but multi-year timeline	Very low	High (if achieved)	Low	8/25

Alternative	Cost	Behavior Change Mechanism	Measurability	Persistence	Feasibility for Team	Score
One-time cleanup event	Medium	Addresses symptoms, not sources	Low	Very low	Medium	5/25
School-only education	Low	Misses primary decision-makers	Medium	Low	High	10/25
Single-channel door hangers	Low	Reaches decision-makers	Medium	Low	High	12/25
Four-prong model (selected)	Medium	Multi-modal; addresses all drivers	High	High	High	22/25

Owner: Project Lead, with all team members contributing to the alternatives discussion at a dedicated October team meeting.

Section 3: Best Solution (20 points)

Judges score the logical connection between the selected solution and the specific problem. Connect each prong explicitly to a behavior-change mechanism:

- **Drain markers → Visual cue at point of decision.** Every resident who approaches a marked drain with a lawn mower or fertilizer spreader sees the consequence before acting.
- **Door hangers → Information delivery to the household decision-maker.** Targets the adult who controls fertilizer application, pet management, and debris management.
- **Classroom sessions → Trusted messenger channel.** A child who tells their parent about the lagoon connection is more credible than a piece of mail.
- **Water quality monitoring → Scientific accountability.** Creates an independent measurement of whether the campaign is actually changing the receiving water, not just household awareness.

Owner: Project Lead and Science Lead jointly.

Section 4: Implementation (20 points)

Judges verify that the team actually did what they said they would. Dated, specific field evidence. This is where your documentation system produces its return.

Every drain log entry, every distribution session log, every classroom session record, every water quality data sheet — all of it is implementation evidence. Team meeting minutes demonstrate collaborative planning. Photographs show execution in progress.

Owner: All five roles contribute their records. Project Lead compiles.

Section 5: Community Impact (15 points)

Judges score measurable, honest evidence. Specific numbers, clearly sourced.

- Q1 knowledge change: [X]% correct before, [Y]% correct after, across [N] survey respondents
- Water quality comparison: honest reporting per Section 9 framing
- Reach: [N] drains, [N] households, [N] students, [N] classroom sessions
- Q5 (Wave 2): percentage who report seeing the drain markers

Do not round up or estimate. 52 drains is more credible than "approximately 50." 11% knowledge change is more credible than "over 10%." Specificity signals integrity.

Owner: Community Outreach Lead (surveys) and Science Lead (water quality). Project Lead integrates.

Section 6: Sustainability (5 points)

Written evidence that the project continues after the team. Institutional commitments in writing.

- City email confirming receipt of drain database and incorporation into marking program
- HOA written acknowledgment of continued community education partnership
- Physical drain markers lasting 3–5 years
- Protocol published on project website for community replication

Owner: Project Lead.

Writing the Proposal — Monthly Schedule

Month	Section	Prerequisites
October 2026	Section 1 — Problem Identification	Pre-survey baseline data collected
October 2026	Section 2 — Alternatives Analysis	Dedicated team meeting for alternatives discussion
November 2026	Section 3 — Best Solution	All four prongs active in the field
December 2026	Section 4 — Implementation (partial)	Drain marking complete, Wave 1 distributed, sessions underway
January 2027	Section 5 — Community Impact (partial)	Post-survey launched, Round 2 water quality logged
February 2027	All sections finalized	All field work complete, Round 3 tested, city handoff scheduled
Late February 2027	Final review and submission	

The Digital Portfolio

Due approximately March 22, 2027. Includes a 3–5 minute project video, supporting documentation, and evidence of community impact.

The video: Plan and film throughout the year — do not try to film everything in March. Capture B-roll during drain marking (July), distribution evenings (July and November), classroom sessions (September–October), and water quality testing (all three rounds). Final video assembly in March is editing existing footage, not starting from scratch.

The Oral Presentation (If Advancing to Florida Finals)

Presentation is typically 10–15 minutes with Q&A. Assign specific sections to specific team members before practice begins.

- Project Lead: Introduction, problem identification, alternatives analysis
- Field Operations Lead: Drain marking, city partnership
- Community Outreach Lead: Hanger design, survey methodology, knowledge change results
- Education Lead: Classroom sessions, student impact
- Science Lead: Water quality methodology, site selection, data results

Practice the full presentation at least 5 times as a team. Record one session on video and review it together. Every team member must know every other member's section well enough to deliver it if needed. Know your numbers without notes — judges will ask follow-up questions on data.

Section 12: HOSA Community Awareness

Overview

HOSA Community Awareness is a portfolio-based competition documenting a public health awareness campaign. It is relevant because the stormwater-to-lagoon pathway is also a documented human health exposure pathway: nitrogen loading drives cyanobacterial blooms, which produce cyanotoxins, which cause respiratory irritation, skin conditions, and

gastrointestinal illness in people who contact affected water or aerosols near bloom events.

H4 status: Open. HOSA does not require citizenship. Health Science course enrollment required — confirm with your HOSA advisor before school starts.

Critical Dates

Date	Action
December 1, 2026	HOSA membership dues — pay first, before anything else that day
Winter/Spring 2027	Regional CE Conference (Brevard County — get date from advisor)
April 15–18, 2027	Florida SLC — Marriott Orlando World Center (CONFIRMED)
June 23–26, 2027	HOSA ILC — Baltimore Convention Center (CONFIRMED)

Health Framing — Every Section Must Connect to Human Health

HOSA judges score public health interventions. Every section of the portfolio must explicitly connect the project to human health outcomes.

The correct chain: Residential nitrogen → stormwater runoff → lagoon nitrogen loading → cyanobacterial bloom → cyanotoxin production → skin irritation, respiratory symptoms, gastrointestinal illness in residents who contact affected water, breathe aerosols, or consume contaminated seafood. Florida Department of Health issues documented advisory notices restricting contact during bloom events in Brevard County. Clearing the Way reduces the nitrogen that initiates this chain at Step 1.

Portfolio — Build Monthly, Not at the End

Section	Content	When to Build
1 — Project Overview	Health problem, target population, campaign rationale, public health connection	September 2026
2 — Activity Documentation	Every activity with dated photos, attendance counts, partner letters	October 2026 (update monthly)
3 — Knowledge Change Evidence	Pre/post survey methodology and Q1 results	December–January (after Wave 2)
4 — Water Quality Data	Site descriptions, round summaries, parameter comparison	January–February 2027
5 — Partner Letters	All signed letters from city, teachers, HOA	Collected throughout year
6 — Sustainability	City handoff acknowledgment, HOA commitment	February–March 2027

Section 13: Earth Prize

Overview

Earth Prize is an international environmental competition for students ages 13–19. The North America regional winner receives \$12,500. The competition

runs in two phases: a written Fresh Idea submission, then a pitch video and live virtual Q&A for finalists.

H4 status: Fully international. No citizenship or residency requirement of any kind.

Register September 1, 2026 at theearthprize.org. Join the Earth Prize Discord community immediately after registering — mentors provide feedback that has historically produced stronger Final Idea submissions. Registration closes January 10, 2027.

What the Fresh Idea Needs

Environmental significance: Cite the published data. The IRL is an EPA National Estuary Program. 60% seagrass loss is published and widely cited. This dimension is won automatically with proper sourcing.

Uniqueness: Most North America competitors submit technology concepts. Clearing the Way is none of that. "No technology required. No capital equipment. No university partnership. Just direct community education with field-tested measurement, at under \$1.00 per household." This framing is genuinely unusual in the Earth Prize application pool.

Feasibility: Describe actual obstacles encountered, not theoretical ones. Concrete problem-navigation stories are proof of feasibility.

Community impact: Lead with cost-per-household. Total budget divided by households reached = approximately \$0.60–\$0.80 per household. This is a remarkable unit economics figure for any behavior-change campaign.

Replicability: The protocol — hanger design, survey questions, GPS logging system, test kit list, city handoff template — is published on the project website. Any coastal community can download and deploy it. This is the scalability argument Earth Prize judges are specifically scoring.

When to write it: January 2027, after all field data is collected. Four sessions of 2 hours each. Use the IYEC essay (Section 15) as the compressed core — the Fresh Idea is approximately 3× longer and adds the replication argument and city handoff narrative.

Section 14: Action for Nature Eco-Heroes

Overview

Action for Nature International Young Eco-Heroes awards up to \$1,000 cash per winner. Online application form.

H4 status: Fully international. Open to ages 8–16 from any country.

AGE CEILING: Every applicant must be 16 or younger on February 28, 2027. Check each team member's age immediately. Any member who will be 17 or older on that date is ineligible this cycle.

Deadline: February 28, 2027.

Application Fields

Field	What to Write
Project goals	"To reduce residential nitrogen loading to the Indian River Lagoon by marking [N] storm drains in two Melbourne, FL neighborhoods and educating the households above them about the direct connection between their streets and the lagoon."

Field	What to Write
Methods	Four bullet points, one per prong. Two sentences each: what you did, how you documented it.
Outcomes/Impact	Specific numbers only: drains marked, households reached, Q1 knowledge change %, students in sessions, water quality rounds.
Photos	Two dated photos: one before/after drain pair, one classroom session.
Duration	State start date (June 2026) and that the project is ongoing.
Why self-initiated	"No teacher assigned this project. The team founded it before 10th grade and initiated all partnerships, logistics, and funding independently."

Write the application in January 2027 after the Earth Prize Fresh Idea draft is complete. Most language adapts directly. Time to write: 3–4 hours.

Section 15: IYEC 500-Word Essay

Why Enter

The International Youth Environmental Challenge (IYEC) is a small but legitimate global competition. The reason to enter is the discipline, not the prize: writing a 500-word essay about this project forces the team to identify the single most important thing about what they did. The sentences that

survive 500 words become the opening of the Earth Prize Fresh Idea, the Eco-Heroes application, and the CmPS proposal introduction.

H4 status: Open. Global, under 18, no citizenship requirement.

Deadline: January 15, 2027. Write December 26–28 during winter break.

Structure for 500 Words

- **Paragraph 1 — The problem (60 words):** Specific problem with two statistics.
- **Paragraph 2 — The information gap (80 words):** Most residents don't know. Pre-survey Q1 data: only [X]% correct before the campaign.
- **Paragraph 3 — The solution (150 words):** Four prongs. One sentence per prong. Concrete terms only.
- **Paragraph 4 — The results (80 words):** Knowledge change %, drains, households, students, water quality comparison.
- **Paragraph 5 — Scale (80 words):** Protocol published free on project website. Any coastal community can replicate it for under \$400.
- **Closing (50 words):** Why this matters beyond your neighborhood.

Write the rough draft in one sitting. Edit twice. Submit by January 15.

Section 16: Other Competitions to Consider

NatGeo Slingshot Challenge

What: 60-second video competition for ages 13–18. Up to \$10,000 in conservation funding and NatGeo platform recognition. **H4 status:** Open.

When: Deadline typically November–December. Verify at natgeo.com/slingshot by September 1, 2026.

The 60-second rule is absolute. Over 60 seconds means automatic disqualification. Have five people who have never heard of the project watch the video before submission. If they cannot summarize what was done and why it matters in one sentence, reshoot.

Recommended if any team member has video editing experience. Film in August during summer when the team has full days and lagoon visuals are at peak.

Brower Youth Award

What: Six winners per year across North America, ages 13–22. Prize: \$3,000 cash, a professional mini-documentary, a leadership retreat in Berkeley. **H4 status:** Open — "Young people living in North America."

Recommendation: Apply in Year 2 when the team has a completed city handoff with written documentation. The Brower committee looks for projects that have created permanent institutional accountability. The city incorporating the team's drain database into its official stormwater program is that story.

Fairchild Challenge

What: A year-long multi-challenge Florida environmental competition. Students submit individual entries across community action, photography, creative writing, scientific report, and digital media categories. Schools accumulate points. **H4 status:** Open — Florida school enrollment is the only requirement.

Contact fairchildchallenge.org in August–September 2026 to confirm West Shore is registered for the 2026–2027 cycle.

Enter every challenge with a reasonable fit: photography (before/after drain photos), community action (campaign narrative), scientific report (water quality data), digital media (awareness video). Each entry earns your school points toward the year-end recognition ceremony at Fairchild's Coral Gables campus.

GENIUS Olympiad — Business Category

What: International competition at Rochester Institute of Technology. The Business category accepts environmental nonprofit and social enterprise plans. Finalists present in-person at RIT in June. Selected entries are published in a peer-reviewed journal. **H4 status:** Open — fully international.

If pursuing: Begin the 8-section business plan in September 2026, one section per month. This is the most documentation-intensive optional competition. Application typically due March 1. Best suited for the team member who enjoys business and systems thinking.

DECA Community Awareness Project

What: DECA competitive event for students documenting a community awareness campaign. Individual or team of up to 3. Requires DECA membership through the school. **H4 status:** Open — DECA membership is the only requirement.

If pursuing: Join DECA in September 2026. DECA ICDC 2027 is confirmed in Atlanta, GA (April 25–29) for teams advancing from Florida SLC.

Section 17: The Application Writing System

Write Once, Adapt Many Times

The team should not rewrite from scratch for each application. The Project Lead maintains a "Master Paragraph Library" — a Google Doc with eight reusable paragraph blocks. Each block is written once and adapted for each application's framing.

The 8 Master Paragraphs

Block	Write In	Reuse In
The IRL in 100 words	October 2026 (CmPS Section 1)	All applications — opening context
The drain-to-lagoon mechanism	October 2026	HOSA health rationale, Earth Prize, Eco-Heroes, IYEC
The awareness gap — pre-survey data	December 2026 (after baseline analysis)	All applications — the "why needed" evidence
The four-prong solution	January 2027 (Earth Prize)	CmPS Section 4, HOSA narrative, Eco-Heroes methods
The metrics paragraph	February 2027 (after all data)	HOSA evidence, Earth Prize, Eco-Heroes outcomes
The knowledge change result	February 2027	Headline number in every application
The city handoff — sustainability	February–March 2027	CmPS Section 6, HOSA sustainability, Earth Prize scalability

Block	Write In	Reuse In
Why we started — team story	Eco-Heroes application	Earth Prize hook, HOSA intro

Writing Schedule

Order	Application	When	Hours
1st	IYEC (500 words)	December 26–28, 2026	3
2nd	Earth Prize Fresh Idea	January 2027	8
3rd	Eco-Heroes application	January 2027	3–4
4th (ongoing)	CmPS Proposal	October 2026 – February 2027	12–15
5th (ongoing)	HOSA Portfolio	September 2026 – March 2027	10–12

Common Mistakes to Avoid

Writing in future tense about planned activities. Every competition section about methods must be past tense about completed work. Write applications after activities happen, not before.

Vague numbers. "Hundreds of households" is weaker than "342 households." "Many students" is weaker than "127 students across 5 sessions." Count everything. Report the actual number.

Overstating water quality results. If the data does not show a statistically clear improvement, say so with the honest framing from Section 9. Judges know the ecological timelines.

Assembling portfolios at the deadline. CmPS and HOSA portfolios are built over months. Week-before assembly produces missing photos, missing letters, and records that cannot be reconstructed. Document in real time, build monthly.

PART 4 — OPERATIONS

Section 18: Week-by-Week Timeline

Role labels: PL = Project Lead · FO = Field Operations Lead · CO = Community Outreach Lead · EL = Education Lead · SL = Science Lead · ALL = all team members · PAR = parent needed

SUMMER 2026

June Week 1 (June 1–7)

Task	Role
Hold first team meeting — assign all roles, review this guide in full	ALL

Task	Role
Create shared Google Drive folder structure with all subfolders	PL
Create master Google Sheets log with all tabs (Drains, Distribution, Sessions, WQ, Survey)	CO + SL
Create shared Google Photos album, grant access to all	FO
Email Melbourne Public Works requesting stormwater drainage maps for both target zones	PL
Identify 3 target schools for classroom sessions (initial list only)	EL

June Week 2 (June 8–14)

Task	Role
Receive drainage maps — follow up by phone if not received by June 9	PL
Identify Zone A and Zone B boundaries from maps	FO + SL
GPS-survey all streets in both zones, logging every drain location	FO
Photograph drain infrastructure before any marking (zone condition baseline)	FO

Task	Role
Identify Campaign Sites A and B and Control Site for water testing — GPS all three	SL
Build campaign zone map in Google Maps My Maps, pinning every drain	FO
Begin Wave 1 hanger design in Canva — pre-survey QR code goes at top of folded panel first	CO
Set up pre-survey in Google Forms — copy Q1–Q4 from Section 7 — generate QR code	CO

June Week 3 (June 15–21) — MOST CRITICAL WEEK

Task	Role
BASELINE WATER QUALITY TESTING — ROUND 1 — ALL THREE SITES	SL (leads) + PAR (drives)
Log all Round 1 data in spreadsheet before leaving each site — photograph data sheet	SL
Test-scan pre-survey QR code on two different phones to confirm it works	CO
Request quotes from 2–3 local print shops for 440 Wave 1 hangers	CO

Task	Role
Order custom vinyl drain stickers from StickerMule.com (if city stickers not yet provided)	FO

June Week 4 (June 22–28)

Task	Role
Print Wave 1 door hangers — pick up from print shop	CO + PAR
Hold second team meeting — confirm July distribution schedule and parent availability	ALL
Confirm parent driver availability for 3 July distribution evenings	PL

July Week 1 (July 1–7)

Task	Role
Register for FPS CmPS at futureproblemsolving.org/florida — July 1, register the day it opens	PL
Begin Wave 1 door hanger distribution — Evening 1 — assign streets before leaving	CO (leads) + ALL + PAR
Begin monitoring pre-survey responses — check every 3 days	CO

July Weeks 2–3 (July 8–21)

Task	Role
Drain marking Sessions 1–3 — Zone A — Saturday mornings	FO (leads) + rotating team members + PAR
Wave 1 distribution — Evenings 2 and 3 — complete Zone A and Zone B coverage	CO (leads) + ALL + PAR
Monitor pre-survey response rate — if below 15 after two weeks, initiate door-knocking	CO
Email science teachers at 3 target schools requesting fall classroom sessions	EL

July Weeks 4–5 (July 22 – August 4)

Task	Role
Drain marking Sessions 4–6 — Zone B	FO (leads) + rotating team members + PAR
Monthly team meeting — produce minutes	ALL
Launch project website — Google Sites — mission, campaign zone map, contact form	PL

August (Pre-School, August 1–20)

Task	Role
Complete all remaining drain marking — all drains done before school starts	FO
Calendar September 1 for Earth Prize registration — do it the day it opens	PL
Follow up with any teachers who have not responded to classroom session requests	EL
Meet with FPS coach — confirm CmPS registration, get proposal deadline in writing	PL
Meet with HOSA advisor — confirm Community Awareness event, confirm dues deadline	EL
Confirm Fairchild Challenge school registration at fairchildchallenge.org	PL
Team debrief meeting — review summer accomplishments, distribute school-year schedule	ALL

SCHOOL YEAR 2026–2027

September 2026

Task	Role	Notes
EARTH PRIZE REGISTRATION — September 1 — go to theearthprize.org immediately	PL	Join Discord immediately after registering
Deliver Classroom Sessions 1–2	EL (leads) + 1 member attends each	Get teacher letter signed ON THE SPOT
Begin CmPS Section 1 — Problem Identification — first draft	PL + CO	Pre-survey baseline needed
Begin HOSA Portfolio Section 1 — Project Overview and Health Framing	EL	Focus on public health connection
Monthly team meeting	ALL	Produce minutes

October 2026

Task	Role	Notes
ROUND 2 WATER QUALITY TESTING — within 48 hrs of 0.5"+ rainfall	SL (leads) + PAR (drives)	Mid-campaign snapshot

Task	Role	Notes
Deliver Classroom Sessions 3–4	EL (leads) + 1 member attends each	Get teacher letter on the spot
Continue CmPS Section 2 — Alternatives Analysis	PL + ALL	Hold dedicated alternatives discussion meeting
Build HOSA Portfolio Section 2 — Activity Documentation	ALL	Dated photos from all activities so far
Monthly team meeting	ALL	Produce minutes

November 2026

Task	Role	Notes
WAVE 2 DOOR HANGER DISTRIBUTION — 3 EVENINGS — post-survey QR at top	CO (leads) + ALL + PAR	Same streets as Wave 1
Log every Wave 2 session same evening	CO	Full session records
Deliver Classroom Sessions 5–6 if scheduled	EL	
Continue CmPS Section 3 — Best Solution	PL	Four-prong mechanism of change
Build HOSA Portfolio Section 3 — Knowledge Change Evidence	CO	Wave 2 post-survey launching now — explain methodology

Task	Role	Notes
Email HOA president requesting 10-minute presentation slot in January or February	PL	
Monthly team meeting	ALL	Produce minutes

December 2026

Task	Role	Notes
PAY HOSA DUES — December 1 — first task of December	EL	No exceptions
Begin calculating pre/post survey results — Q1 correct rate before vs. after	CO	
Continue CmPS Section 4 — Implementation — each role contributes their records	ALL	
Write IYEC 500-word essay — December 26–28 — submit by January 15	PL + CO	Use structure from Section 15
Monthly team meeting	ALL	Produce minutes

January 2027

Task	Role	Notes
Submit IYEC essay — January 15	PL	
Earth Prize registration closes — January 10 — must already be registered	PL	
ROUND 3 WATER QUALITY TESTING	SL (leads) + PAR (drives)	Final round. Post-campaign.
Write Earth Prize Fresh Idea — 4 sessions × 2 hours	PL + SL + CO	
Write Eco-Heroes application — 3–4 hours — adapt Earth Prize paragraphs	CO	
Monthly team meeting	ALL	Produce minutes

February 2027

Task	Role	Notes
Submit Eco-Heroes — February 28	CO	
Submit Earth Prize Fresh Idea — ~February 28	PL	Verify exact date

Task	Role	Notes
Submit CmPS Proposal — ~February 27	PL	Verify with FPS coach
Compile city handoff package	FO + CO + SL	All documents formatted
City handoff meeting — present package, get written email confirmation	PL + FO + SL + PAR	
HOA presentation — 10-minute slot	PL + EL + PAR	Get written acknowledgment
Monthly team meeting	ALL	Produce minutes

March 2027

Task	Role	Notes
CmPS Digital Portfolio + Video due ~March 22	ALL	Video assembled from B-roll captured all year
CmPS Florida Affiliate Finals ~March 23–26 — attend in person	ALL + PAR	Know your numbers without notes
Finalize HOSA portfolio for regional competition	EL + ALL	
HOSA Regional CE Conference — Brevard County	ALL + PAR	Get exact date from advisor

April 2027

Task	Role	Notes
HOSA Florida SLC — April 15–18, Orlando	Advancing members + PAR	Marriott Orlando World Center. Plan travel by February.
Begin Year 1 Annual Report draft	PL	Summary of everything accomplished — for website and college applications

Section 19: Budget & Finances

Overview

Clearing the Way operates as a school club or project-level initiative — no nonprofit filing fees. Costs are covered through team member contributions, parent support, and school club funds where available.

Full Budget

Item	Cost	When
Wave 1 door hangers — 440 units (400 + 10% overage)	\$45–60	Late June
Wave 2 door hangers — 440 units	\$45–60	Late October

Item	Cost	When
Custom vinyl drain stickers — 100-pack, StickerMule	\$25–30	July (if city stickers not free)
Stencil (if using paint method)	\$15–25	July
Spray paint — Rust-Oleum, 2 cans	\$12	July (if using stencil)
Alcohol wipes for drain surface prep	\$5	July
API Freshwater Master Test Kit	\$35	Before Round 1
LaMotte phosphate test strips	\$18	Before Round 1
Waterproof digital thermometer	\$10	Before Round 1
Classroom demo supplies (jars, food coloring, soil, oil)	\$8–12	Before Session 1
Take-home cards — 200 total for 4–6 sessions	\$20–30	Before Session 1
Website domain (optional — Google Sites free without custom domain)	\$12/year	July
TOTAL PROJECT COSTS	~\$250–\$320	

Funding Sources

Source	Amount	Timing
Team member contributions (5 members)	\$50 each = \$250	As needed
School club budget (if applicable)	Request at school year start	Before June
Parent contributions	As offered	As needed

Cost Reductions

- Melbourne Public Works may provide "Drains to Lagoon" stickers free — request in your first email. If confirmed, remove the sticker/stencil budget line entirely.
 - Schools may print take-home cards at no cost with a teacher sponsor's assistance.
 - Wave 2 hanger quantity can be reduced if Wave 1 response data shows low engagement on specific streets.
-

Section 20: When Things Go Wrong — 12 Scenarios

Scenario 1 — Pre-survey response rate is under 15 after two weeks

Have two team members knock on 10 doors per evening in the campaign zone for one week. Script: "Hi, we're West Shore students running a lagoon protection project. Can I get 60 seconds of your time for a quick survey?" Face-to-face student asks produce a significantly higher yes rate than a QR code. 3–4 evenings of door-knocking gets you to 20–25 responses.

Scenario 2 — The city does not respond to emails

Call Melbourne Public Works directly: (321) 674-5752. "We emailed twice about a drain marking partnership for a student project. We just need 15 minutes with whoever handles the stormwater program." Phone calls produce responses that emails do not. You need one yes from one person.

Scenario 3 — A teacher says no to a classroom session

Ask for a colleague referral — a referral doubles your acceptance rate with the referred teacher. Alternatively, email the school PTA chair and request 10 minutes at a PTA meeting. Thirty parents at a PTA meeting are an excellent audience — they are the exact household decision-makers the campaign targets.

Scenario 4 — Water quality data shows no change

Report it exactly as it is using the language from Section 9. No change in a single year is scientifically expected and not a competition failure. Judges who evaluate water quality projects know the ecological timelines. Your pre/post survey knowledge change data is the primary evidence of impact.

Scenario 5 — A team member is not contributing

The Project Lead schedules a one-on-one conversation — not a group discussion — with that member. Review role responsibilities and ask directly: "What's making it hard to complete [specific task]?" If the issue is temporary,

redistribute temporarily and set a check-in date. If persistent, the team meets and formally reassigns responsibilities — documenting the change in meeting minutes.

Scenario 6 — A major exam week falls during a key field activity

Reschedule the activity. Exam performance is not sacrificed for a competition project. Most activities have a 2–3 week window of flexibility. The only exception is the Round 1 baseline water quality test, which must happen before Wave 1 distribution. Every other activity can flex.

Scenario 7 — The HOA president does not respond

Attend a public HOA meeting, which is open to any resident. Most HOA meetings include a brief public comment period. At that moment, introduce the team and the project and offer to present at the next meeting. An in-person introduction at a public meeting produces results that emails do not.

Scenario 8 — The print shop cannot meet the hanger deadline

Online printing services (GotPrint, Vistaprint, Office Depot online) offer door hanger printing with 3–5 day turnaround. Keep the Canva design file saved as a print-ready PDF — it can be uploaded directly to any online printer with no reformatting.

Scenario 9 — Team morale drops in November–January

This is the highest-burnout period: Wave 2 distribution is tiring, competition writing is ramping up, and school demands are high. At the November or December meeting, acknowledge this directly. Then review what the team has actually completed — drain count, households reached, sessions delivered. The evidence is more motivating than any encouragement. Break every remaining task into the smallest possible unit: not "write CmPS Section 4" but "write 200 words about drain marking Session 2 by Friday."

Scenario 10 — The city does not follow through on the handoff

You need one written email from a city address — not a formal letter, not a verbal commitment. If no written acknowledgment arrives within 2 weeks of the meeting, email the contact: "Following up — could you send a quick email confirming you received the database? Even one sentence works. My competition deadline is [date]." Most people respond when given the exact format and a deadline.

Scenario 11 — The CmPS video comes out poorly

Prioritize: (1) good audio — use a lapel or external microphone, or film in a quiet room. (2) Clear visuals — before/after drain photos, jar demonstration, water testing. (3) Data on screen — key numbers as text overlays. A sincere team presenting real data in an imperfect video is more compelling than a polished video with vague claims.

Scenario 12 — A competition deadline changes from this guide

Set a calendar reminder for September 1, 2026 to independently verify every competition deadline from the official website. This guide uses verified dates as of the planning period, but competition deadlines change annually. Never rely on any planning document for a deadline you have not personally verified from the official source within the current cycle.

Section 21: Quick Reference Cards

Card A — Team Roles

Role	Owner	Primary Deliverable
Project Lead / CmPS	_____	CmPS Proposal + city handoff
Field Operations	_____	Drain database + photo archive
Community Outreach	_____	Hangers + surveys + distribution logs
Education Lead	_____	Classroom sessions + teacher letters
Science & Data	_____	Water quality data + analysis

Card B — Never Miss These Dates

Date	Action	Consequence
June 15, 2026	ROUND 1 BASELINE WATER QUALITY TEST	Baseline permanently lost if outreach begins first
July 1, 2026	FPS CmPS registration opens — register today	Late registration may miss the Florida cycle
September 1, 2026	Earth Prize registration opens — register today, join Discord	Registration closes January 10
December 1, 2026	HOSA dues — pay first thing	Cannot compete in any HOSA event this year
January 10, 2027	Earth Prize registration closes	Must already be registered
January 15, 2027	IYEC essay deadline	Write December 26–28
February 28, 2027	Eco-Heroes + Earth Prize Fresh Idea	Both due same day
~February 27, 2027	CmPS Proposal	Verify with FPS coach
April 15–18, 2027	HOSA Florida SLC, Orlando	Plan travel by February

Card C — Numbers Every Team Member Knows Without Notes

Statistic	Value
IRL species count	4,300+

Statistic	Value
Seagrass lost since 2011	60%
Annual economic impact	~\$430M/year
Drains marked (target)	50–75
Households reached (target)	300–500
Classroom sessions	4–6
Students reached	80–150
Water quality rounds	3
Water quality sites	3 (2 campaign + 1 control)
Cost per household (estimate)	~\$0.60–\$0.80
Q1 knowledge change result	[Fill in after post-survey]

Card D — The 3-Minute Field Log (Do This Before You Leave Every Site)

1. Take the "after" photo. Confirm it is in focus.
2. Open Google Sheets on your phone. Add one complete row.
3. Fill in: Date · Activity Type · Location/ID · Team Members Present · Photo Filenames · Notes
4. Set an alarm for tonight: "RENAME PHOTOS"
5. When alarm fires: rename all photos using
YYYYMMDD_type_identifier.jpg
6. Upload renamed photos to the shared Google Photos album.

Every time. Without exception.

Card E — One-Sentence Pitches by Competition

Competition	Opening Sentence
CmPS	"We identified a quantified information gap — only [X]% of households above our study zone's storm drains knew those drains flow directly to the lagoon — and built a four-prong intervention to close it."
HOSA	"We interrupted the stormwater-to-cyanotoxin exposure pathway at Step 1 by changing the residential behaviors that drive nitrogen loading into the Indian River Lagoon."
Earth Prize	"We built and field-tested a replicable residential stormwater awareness protocol for under \$400 — and published the complete guide for any coastal community to deploy."
Eco-Heroes	"We marked [N] storm drains, educated [N] households, brought the lagoon science into classrooms, measured the water quality results, and handed the program to the city — starting before 10th grade."
IYEC	"In a 156-mile estuary home to 4,300 species, we proved that the most powerful intervention is also the

Competition	Opening Sentence
	cheapest: telling people what their storm drains actually connect to."

Card F — H4 Visa Eligibility Summary

Competition	H4 Status
CmPS (FPS Florida)	OPEN — Florida school enrollment only
HOSA Community Awareness	OPEN — Health Science enrollment only
Earth Prize	OPEN — Fully international, no residency requirement
Action for Nature Eco-Heroes	OPEN — International, ages 8–16, any country
IYEC	OPEN — Global, under 18, no citizenship requirement
NatGeo Slingshot	OPEN — Ages 13–18 worldwide
Fairchild Challenge	OPEN — Florida school enrollment only
DECA Community Awareness	OPEN — DECA membership through school only

Not eligible: Gloria Barron Prize (requires SSN + permanent residency), Riley's Way C4K (requires US citizen or DACA), VFW Voice of Democracy (requires US citizenship).

Card G — Parent Monthly Update Template

Clearing the Way — [Month] Update

What happened this month: [2–3 sentences]

What's coming next month: [2–3 sentences]

Parent involvement needed: [Specific dates, what is needed, who should confirm]

No action needed on: [Anything they might wonder about but don't need to do]

Keep every update under 150 words. Parents who receive short, clear updates stay engaged. Parents who receive long updates stop reading them.

Card H — Application Paragraph Reuse Map

FIELD LOG + SURVEY DATA + WATER QUALITY DATA

|



IYEC ESSAY — December 26–28

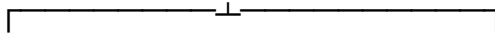
500 words · January 15

|



EARTH PRIZE FRESH IDEA — January 2027

Expanded ×3 · Due ~February 28



ECO-HEROES — January 2027 CmPS PROPOSAL — Oct 2026–Feb 2027

Due February 28

Built section by section monthly



HOSA PORTFOLIO — Sept 2026–Mar 2027

Same data, public health framing

A Final Word

This project is not principally about winning competitions. Competitions are the external recognition mechanism for something that matters regardless of any award: families in Melbourne are going to learn a true fact about where their water goes, and some of them are going to act on it. Storm drains in two neighborhoods will carry markers communicating that fact for the next

several years. Water quality data will exist in a spreadsheet — honest, dated, and collected by this team — that did not exist before.

The applications write themselves when the field work is real. Do the work. Document in the moment. Let the evidence carry the applications.

The sentence every judge will remember:

"We built something real, measured it honestly, and handed it to the city so it runs without us."

That sentence is only true if the team actually does those things. Start June 1.

*Clearing the Way Initiative · West Shore Jr./Sr. High School · Melbourne,
Florida · Indian River Lagoon · Summer 2026*